

Introduction to Managed Service for Apache Spark

Welcome to Managed Service for Apache Spark! You'll explore what makes it such a powerful tool and how you can implement it in order to both save time and increase the reliability of your data.

Let's begin with an example.



Maybe you work for a major online retailer that sells more than five-hundred products and sources them from about 2,000 suppliers. Obviously, this adds up to A LOT of data—both structured and unstructured.

This retailer is working on a strategy to stay competitive in the fast-paced world of online retail. Staying ahead of the competition is a top priority. That's why your company is so focused on building a cutting-edge data system that revolutionizes the way they process data and customer reviews.

Managed Service for Apache Spark

To help achieve this objective, you and your data team utilize Managed Service for Apache Spark to manage the unstructured data. Managed Service for Apache Spark is a fully managed service that maximizes open-source data tools for batch processing, querying, streaming, and machine learning.

One of the benefits of Managed Service for Apache Spark is that you can increase and decrease compute resources based on the project, so then you don't need to guess, under-estimate, or over-estimate resources.

This means your online retailer can dedicate cloud computing resources that focus only on processing customer reviews. And you can have separate resources that focus only on processing sales information.

You also learn that Managed Service for Apache Spark works especially well with Apache Hadoop, a framework that distributes the processing of data across clusters of computers.

This also means that we can scale up to use the power of multiple computers ... even thousands! to process our data in real-time when we need it.

Managed Service for Apache Spark **uses its open-source tools on virtual machines** that **scale up and down as needed**. And it can **install a Hadoop cluster in 90 seconds**.

How it works

Let's learn a bit about how Managed Service for Apache Spark scaling works!

Managed Service for Apache Spark **disaggregates storage and compute services**, so both can be created and terminated as needed. This means that, once you have data to store, your storage will remain. But if you're not actively processing data, the **compute services will terminate**.

For a more specific example, let's go back to our online retailer. So, your company is going to **store** external application logs in cloud storage. Then, the data will be **processed** by Managed Service for Apache Spark, which can **write** it back to Cloud Storage or BigQuery. You can also **send** the data to a data science notebook for analysis or to data scientists who can build and train AI models.

Because storage is separate from processing, your organization will save money by grouping tasks. So, you'll be using exactly the right amount of memory and storage space to meet all your data analysis needs.

Templates

Your company has always stored data across different on-premises and hardware-based servers. You learn that Managed Service for Apache Spark offers pre-built templates that can help you migrate your data from your existing infrastructure into Cloud Storage or BigQuery. Some of the common templates include **Snowflake-To-Cloud Storage**, **Redshift-To-Cloud Storage**, **S3-To-BigQuery**, and **Kafka-to-BigQuery**. This means your databases and workloads in Snowflake, Redshift, or AWS can be automatically transferred into Google Cloud using these templates.

In addition, Managed Service for Apache Spark makes it **easier to manage Hadoop**, and you can **integrate organization-wide security**. Plus, you can **enable data users through integrations**, giving varied permissions to different users.

All right, now that your company has integrated Managed Service for Apache Spark, it's able to provide data on all five-hundred products to all of its vendors. This has given your data team increased visibility into the best products to source and sell!

Best of all, your suppliers receive a smoother, consistent flow of valuable recommendations, enabling each vendor to tap into the waves of user demand like never before. With real-time feedback and actionable insights, suppliers can fine-tune their product descriptions, ensuring they resonate with users on a whole new level!

Dataflow

You also use Dataflow in this project to **create a data pipeline**. Dataflow gives you the option of either **stream or batch processing**. This makes it possible to **stream** data from your various apps into one database. You can also use the batch processing feature to import any data you already stored on your computer.

Cloud Data Fusion

Cloud Data Fusion is the tool you use to build a data pipeline. You can also **create** data integrations, **run** high-volume data pipelines, and **continuously** integrate user data from your applications into your database.

BigQuery

Once you have data from all sources streaming into Cloud Storage, you use BigQuery to **write Sequel queries to join data** and **clean the data** to ensure it's complete, has no duplicates, and will be useful for your project.

Managed Service for Apache Spark

Next, Managed Service for Apache Spark allows you to **use open-source data analytics tools at a large scale**. And you can **apply programming languages and algorithms to data** also at a large scale. With these tools, you're able to move your data to the next phase and start drawing conclusions.

After analyzing the data using the relevant tools, you share your discoveries with the advertising team, highlighting how many users click on ads and what percentage of total gamers that equals. With this information, the advertising team can effectively choose whether to keep using ads to generate revenue or charge users a fee.

You've just experienced a typical business case involving data management and storage. Along the way, you successfully integrated multiple tools to complete a data workflow and find important answers for your stakeholders.

Cloud data professionals are excellent data stewards—and, with the strategies from this example case, you now understand how to use key tools and solutions to PROTECT and

CONTROL all kinds of data!